

Kyle Main

Dallas/Ft. Worth Area | 469-545-3791 | main.kyle87@gmail.com

Senior Security Data Scientist — Anomaly Detection, ML & Security Analytics

Security data scientist, 12 years, building statistical/ML detection systems against massive-scale telemetry — unsupervised clustering, time-series anomaly detection, and large-scale EDA (GCP Dataproc/PySpark) behind 220+ log sources and 2,300+ production detection rules across MITRE ATT&CK. Hands-on GenAI/LLM for security: prompt engineering for false-positive triage/rule generation, GenAI-orchestrated SIEM APIs. M.S. Physics (numerical data analysis & modeling).

DATA SCIENCE & MACHINE LEARNING

Languages & Libraries: Python, SQL, R; Pandas, scikit-learn, NumPy, SciPy, PyTorch

Modeling Techniques: Unsupervised/clustering ML (device behavior clustering on network telemetry), time-series anomaly detection (authentication volume/attempt anomalies, process-chain behaviors), statistical and behavioral detection modeling

Big Data & Cloud ML Platforms: GCP Dataproc/Zeppelin, PySpark/SparkSQL for large-scale EDA; Apache Beam/GCP Dataflow for high-volume batch retrieval; AWS, GCP, Azure cloud security

GenAI / LLM for Security: Prompt engineering for security use cases (false-positive triage, automated detection-content generation); GenAI-driven orchestration of SIEM APIs across customers/platforms; built reusable GenAI "skills" automating detection-rule translation between SIEM syntaxes

SECURITY DETECTION ENGINEERING & DATA PIPELINES

Detection Content: 2,300+ production detection rules (signature, statistical, behavioral, ML-based) across the MITRE ATT&CK matrix; multi-SIEM detection-as-code orchestration (Splunk, Sentinel, Defender, Google SecOps, CrowdStrike, SentinelOne) via native APIs, with GitLab CI/CD, automated testing, and staged rollout

Data Engineering: Pipelines ingesting 220+ unique log sources; 50+ Logstash normalization filters; Common Information Model (CIM) data-dictionary design; Elasticsearch (queries, transforms, Beats, detection rules, API), Splunk

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built an entirely new security data ingestion and analytics platform for DOE/NNSA from the ground up — CrowdStrike, Suricata, and Zeek telemetry into a central Elasticsearch environment, with a UEBA (User and Entity Behavior Analytics) detection layer built on custom data transforms to statistically surface anomalous entity behavior, plus data-quality monitoring/alerting and custom dashboards (DOE/NNSA Security Data Integration project, completed).
- Currently create and manage detection/alerting analytics directly supporting Treasury's Security Operations Center, translating security data patterns into actionable case content for incident response analysts (Treasury SOC / TSSOC, current project).
- Supported data ingestion/quality efforts in an Elasticsearch/Splunk environment for DOE's Continuous Diagnostics and Mitigation (CDM) program (completed).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building signature, statistical, behavioral, and ML-based detection models against massive-scale customer telemetry on a cloud-based big-data platform, translating analytical findings into production risk-detection content.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Built unsupervised clustering models to group network devices by behavioral feature similarity, and time-series anomaly-detection content for entity behaviors — authentication volume/attempt anomalies by country, and parent/child process-chain deviations — as core ML-based detection logic.
- Ran exploratory data analysis at scale on GCP Dataproc compute clusters using Zeppelin notebooks and a home-grown reusable analysis toolkit; wrote PySpark/SparkSQL jobs to load and analyze bucketed data, and a homegrown Apache Beam/GCP Dataflow program to retrieve high-volume historical cold-storage data for customer investigations.
- Created and managed 2,300+ individual detection rules covering most of the MITRE ATT&CK matrix as a very early hire, and owned data engineering for the pipeline feeding them — 220+ ingested log sources, 50+ Logstash normalization filters, and a Common Information Model standardizing fields across all parsed data.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale security log data to develop custom statistical models for DNS-based malware detection/mitigation and anomalous network behavior discovery.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) — Numerical Data Analysis & Modeling, Applied Physics | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox